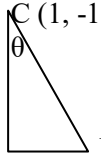
Question			Answer	Marks	AOs		Guidance
12	(i)		C is (1, -1)	B1 [1]	1.1a	Cao	
	(ii)	A	EITHER Substitute $y=\frac{3}{4}x-8$ into the equation of the circle $(x-1)^2+\left(\frac{3}{4}x-8+1\right)^2=25$ $x^2-8x+16=0$ EITHER $(x-4)^2=0$ OR Discriminant = $(-8)^2-4\times1\times16=0$ So the equation has a repeated root so the line is a tangent	M1 M1 A1 A1 [4]	3.1a 1.1a 1.1b 2.2a	**AG** Attempt to eliminate one variable Attempt to expand and collect terms to obtain 3 term quadratic expression A correct 3 term quadratic Clearly argued	
	(ii)	A	OR Substitute $x=\frac{4}{3}y-\frac{32}{3}$ into the equation of the circle $\left(\frac{4}{3}y-\frac{32}{3}-1\right)^2+(y+1)^2=25$ $y^2+10y+25=0$ EITHER $(y+5)^2=0$ OR Discriminant = $10^2-4\times1\times25=0$ So the equation has a repeated root so the line is a tangent	M1 M1 A1 A1 [4]	3.1a 1.1a 1.1b 2.2a	**AG** Attempt to eliminate one variable Attempt to expand and collect terms to obtain 3 term quadratic expression A correct 3 term quadratic Clearly argued	
		B	$x=4$ and $y=-5$ so B is (4, -5)	B1 [1]	1.1a	Cao	

Question	Answer	Marks	AOs		Guidance
(iii)	$\angle CAD = \angle CBD = 90°$ (radius is perpendicular to the tangent)	**B1**	**2.1**	Allow for one or other of these angles	Allow up to B1, B1 for any two of these three pieces of evidence. Allow the final B1 only when the proof is complete and clearly argued.
	Gradient of AC $= \frac{2-(-1)}{5-1} = \frac{3}{4}$				
	Gradient of BC $= \frac{(-1)-(-5)}{1-4} = -\frac{4}{3}$	**B1**	**3.1a**		
	So AC is perpendicular to BC so $\angle ACB = 90°$				
	So ADBC is a rectangle				
	Either AC = BC radius [=5]				
	Or AD = BD equal tangents	**B1**	**2.1**	Complete proof	
	so ADBC is a square.	**[3]**		**AG**	
	$\angle CAD = \angle CBD = 90°$ (radius is perpendicular to the tangent)	**B1**		Allow for one or other of these angles	
	Gradient of AC $= \frac{2-(-1)}{5-1} = \frac{3}{4}$				
	Gradient of BD is $\frac{3}{4}$				
	So AC is parallel to BD So ADBC is a rectangle	**B1**			
	AC = BC = radius	**B1**		Complete proof	
	so ADBC is a square.			**AG**	
	$\angle CAD = 90°$ (radius is perpendicular to the tangent)	**B1**			
	AC = BC radius [=5]	**B1**			
	Gradient of AC $= \frac{2-(-1)}{5-1} = \frac{3}{4}$				
	Equation of AD is $y - 2 = -\frac{4}{3}(x-5)$			Gradient of AD must be found from the coordintes of A and C	
	So coordinates of D are (8, -2)				
	Hence BD = 5 and AD = 5	**B1**		Complete proof	
	So ABCD is a rhombus				

Question	Answer	Marks	AOs		Guidance
(iv)	E is the point (1, -6)	B1	2.1	May be implied	
	EITHER				
	C (1, -1) θ  B (4, -5)	M1	3.1a	Right-angled triangle formed and use of arctan	
	$\theta = \arctan\left(\frac{3}{4}\right) = 0.6435$	A1	3.1a	oe	
	OR				
	$BE = \sqrt{(4-1)^2 + (-5-(-6))^2} = \sqrt{10}$				
	Cosine rule in triangle BCE				
	$\cos BCE = \frac{5^2 + 5^2 - 10}{2 \times 5 \times 5}\left[= \frac{40}{50}\right]$	(M1)		Using distance BC and the cosine rule	
	$\angle BCE = 0.6435...$	(A1)		oe	
	OR				
	M is the midpoint of BE M is (2.5, -5.5)				
	$BM = \sqrt{(4-2.5)^2 + (-5-(-5.5))^2} = \frac{1}{2}\sqrt{10}$				
	$\angle BCM = \arcsin\left(\frac{\frac{1}{2}\sqrt{10}}{5}\right) = 0.32175...$	(M1)		Using trig in triangle BCM or ECM Allow for $\angle BCM$	
	$\angle BCE = 0.6435...$	(A1)		Oe. Must be $\angle BCE$	
	Area sector $= \frac{1}{2}r^2\theta = \frac{1}{2} \times 25 \times 2 \times 0.32175...$	M1dep	1.1a	Using the sector area formula	
	$= 8.04376$	A1 [5]	1.1b	FT their $\angle BCM$	